Bharath Mandha

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Summary

AI/ML engineer with 4+ years building ML systems in finance, healthcare and retail; at Goldman Sachs, focused on retrieval, LangGraph orchestration and LLM evaluation for a production RAG platform.

Experience

Goldman Sachs, AI Engineer — Dallas, TX | Jun 2025 – Present

RAG platform over 3M+ internal financial documents, used weekly by 70+ credit and risk analysts

- Co-designed the retrieval and LangGraph orchestration layers; team-estimated ~40% reduction in credit-memo preparation time
- Built the 500+ query evaluation benchmark, measuring retrieval with Hit Rate and MRR and answer quality with RAGAS, used to compare chunking, top-k and embedding choices; supported production releases maintaining 90%+ benchmark groundedness
- Implemented chunking, metadata filtering, reranking and semantic caching, and tuned FAISS for recall, latency and memory
- Built LangGraph workflows for retrieval, analysis, validation and delegation, with analysts reviewing sensitive decisions
- Added incremental FAISS updates to the team's PySpark and S3 ingestion pipeline, so routine changes skip a full rebuild
- Wrote the FastAPI integration with PII redaction, prompt-injection tests, role-based access and audit traces
- Evaluated quantized and LoRA/QLoRA open models (Hugging Face, PEFT) on quality and cost; the savings didn't justify moving complex credit reasoning off the larger model

Cigna Healthcare, Machine Learning Engineer — Remote, India | Mar 2022 – Aug 2023

30-day hospital readmission risk over 1.5M+ HIPAA-regulated encounters

- Built XGBoost and LightGBM readmission models that reached 0.91 ROC-AUC on a held-out cohort of 200K+ encounters (private data), validated with time-aware, patient-level splits
- Engineered 150+ candidate features from labs, demographics, encounter history and clinical notes, about 100 of them kept in production; handled class imbalance and tuned the decision threshold
- Put training, registration, batch inference and retraining into production (MLflow, Airflow, SageMaker, EKS, Terraform) with PSI drift monitoring and SHAP explanations that helped clinical reviewers prioritize high-risk patients

Deloitte, Associate Data Scientist — Hyderabad, India | Jun 2020 – Feb 2022

- Cut weighted MAPE of SKU- and store-level demand forecasts by ~14% (relative) with gradient-boosted models, benchmarked against time-series baselines on time-based holdouts
- Built PySpark, SQL and BigQuery pipelines over 50M+ transactions
- Raised response on comparable campaigns by ~12% with six customer segments from clustering and cohort analysis

Projects

finance-rag-eval: page retrieval over 360 SEC filings — 2026

- On 100 held-out FinanceBench questions, company and period filtering plus statement routing took BM25 Hit@10 from 0.14 to 0.71. A MiniLM hybrid only tied the best BM25 setup (0.73 vs 0.75) at about 5x the median latency; BM25 serves at 11.6 ms p95

Agentic-Research-Assistant: does a LangGraph review loop beat a single pass? — 2026

- A wider single pass beat the review loop on 150 QASPER test questions (68.0% vs 58.7% judged correct, a baseline added after the first run), and the loop only tied a 5-passage pass (56.0%). The judge matched 35 of 40 hand grades

readmission-capacity: readmission risk at a fixed call capacity — 2026

- On the UCI diabetes extract (101,766 encounters) split by patient, a 10% LightGBM call list reaches 25.0% of 30-day readmissions vs 20.5% for prior-stay ranking and 15.8% for LACE. CI retrains and fails if the numbers move

Skills

LLM and retrieval: RAG, LangGraph, LangChain, FAISS, BM25, hybrid retrieval, reranking, semantic caching, Hugging Face, PEFT, LoRA/QLoRA, quantization

Evaluation: RAGAS (groundedness, faithfulness, answer relevance), Hit Rate, MRR, LLM-as-judge, prompt-injection testing

Machine learning: XGBoost, LightGBM, scikit-learn, SHAP, time-series forecasting, clustering, feature engineering

Languages and infrastructure: Python, SQL, PySpark, BigQuery, AWS (S3, SageMaker, EKS), Kubernetes, Docker, FastAPI, MLflow, Airflow, Terraform, GitHub Actions, ONNX Runtime

Education

University of North Texas, Master of Science in Computer Science — Aug 2023 – May 2025